Chapter 25

Post Cardiac Arrest

A majority of patients with cardiac arrest remain unconscious despite successful restoration of spontaneous circulation. The EEG is useful to evaluate for nonconvulsive seizures and make prognostic estimations. It suppresses immediately following circulatory arrest but early electrographic improvements such as some restoration of physiologic rhythms (typically first 12 hours) is highly predictive of a favorable outcome whereas unfavorable patterns (called highly malignant patterns) are predictive of poor outcomes. However, the EEG alone isn't sufficient to reliably estimate outcomes in this critically ill population and it should be used in combination with other independent markers such as such as clinical examination (myoclonic SE, incomplete recovery of brainstem reflexes), bilateral absence of somatosensory evoked potentials (SSEPs) and/or neuroimaging as part of a multimodal approach [1,2].

Prior EEG based prognostication has been limited by the lack of a standardized classification system, interrater reporting variations, and confounding effects of therapeutic hyperthermia and sedation. Recently, there has been growing consensus on the definitions of electrographic patterns using standardized critical care EEG terminology that may reliably estimate prognosis in comatose patients after cardiac arrest [3].

These patterns are classified based on their prognostic significance as highly malignant, malignant, and benign. A recent multicenter study of routine EEGs of 103 comatose survivors recorded around 48 to 72 hours after cardiac arrest (upon rewarming) concluded that highly malignant patterns were 100% specific and about 50% sensitive for poor outcome (defined as best cerebral performance category score of 3–5 until 180 days). Individual malignant patterns had low specificity and sensitivity to estimate poor prognosis (around 48%) but if two malignant patterns were present then specificity for a poor outcome increased to 96%. Benign EEGs were predictive of favorable prognosis with a poor outcome present in only 1% of those patients. Additionally, therapeutic

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hypothermia and sedation did not significantly affect the pretest probabilities of electrographic patterns in this study. These results were corroborated by another study using continuous EEG showing background suppression and burst-suppression with identical bursts (highly malignant patterns) within 24 hours to be highly predictive of a poor outcome and recordings beyond 24 hours had no additional predictive value [3,4].

Approach to Electrographic Patterns Post Cardiac Arrest

1. Identify the electrographic pattern and understand their prognostic significance.
2. Identify nonconvulsive seizures.

Electrographic Patterns Post Cardiac Arrest

The EEG post cardiac arrest (cerebral anoxic injury) may be classified into one of the following three types:

1. ***Highly Malignant EEG***: There is either suppressed background without discharges (Figure 25.1), suppressed background with continuous periodic discharges (Figure 25.2) or burst-suppression with or without discharges (especially with identical bursts) (Figure 25.3).
2. ***Malignant EEG***: There may be malignant abnormalities of foreground, background, or reactivity. The presence of any two of these is highly specific for a poor prognosis.

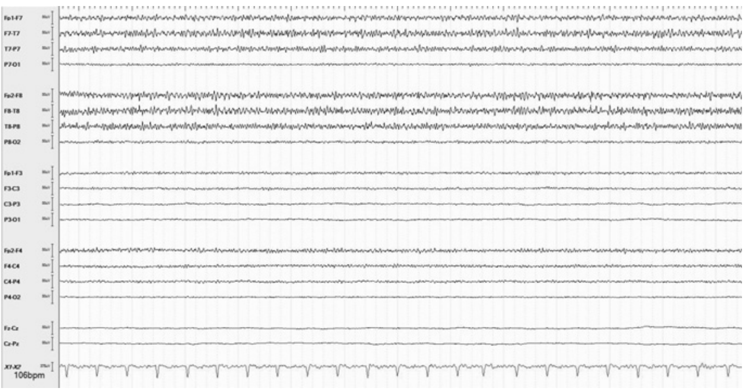

Figure 25.1 Seventy-eight-year-old man with cardiac arrest, suppressed background without discharges.

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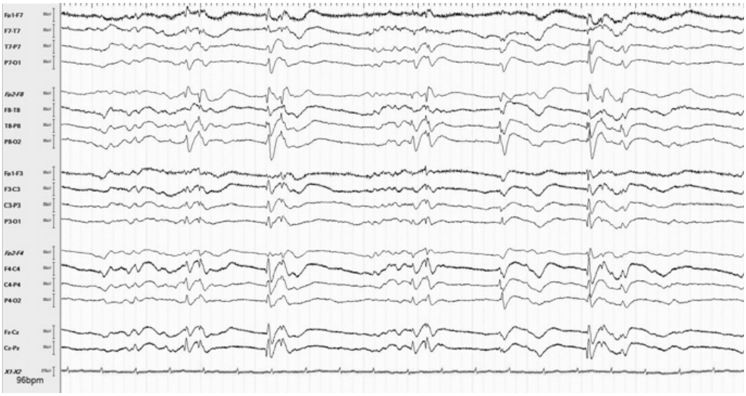

Figure 25.2 Sixty-two-year-old woman with cardiac arrest, suppressed background with continuous generalized periodic discharges.

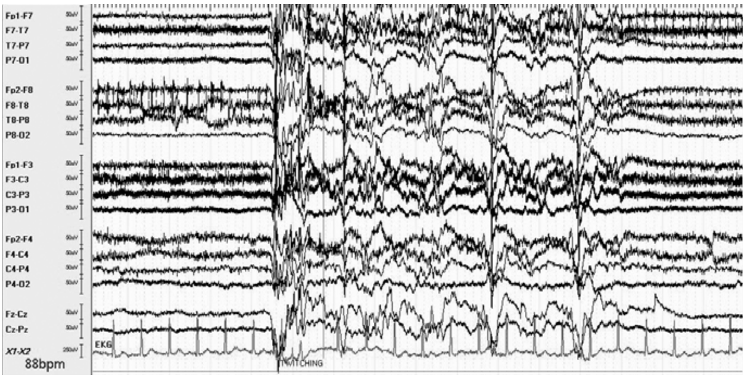

Figure 25.3 Seventy-eight-year-old man with cardiac arrest, burst-suppression (identical bursts) associated with myoclonic status epilepticus (time base of 20 s).

a. foreground: periodic (Figure 25.4) or rhythmic discharges (Figure 25.5) and unequivocal electrographic seizures (ictal patterns)
b. background: low voltage, discontinuity and/or reversals of anterior–posterior gradient (alpha-theta coma as shown in Figure 25.6)
c. loss of background reactivity (or only stimulus-induced discharges)

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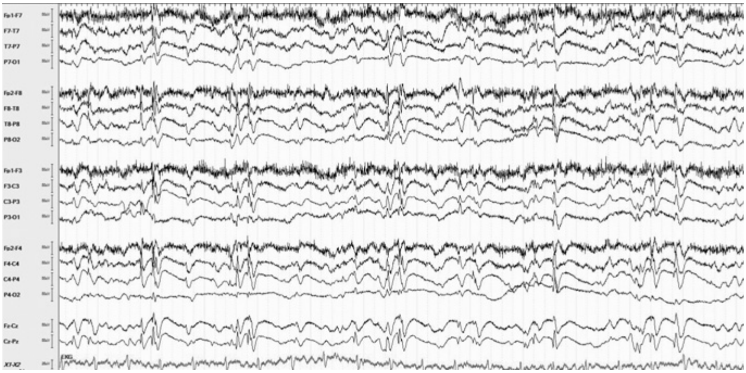

Figure 25.4 Sixty-seven-year-old woman with cardiac arrest, periodic discharges associated with myoclonic status epilepticus.

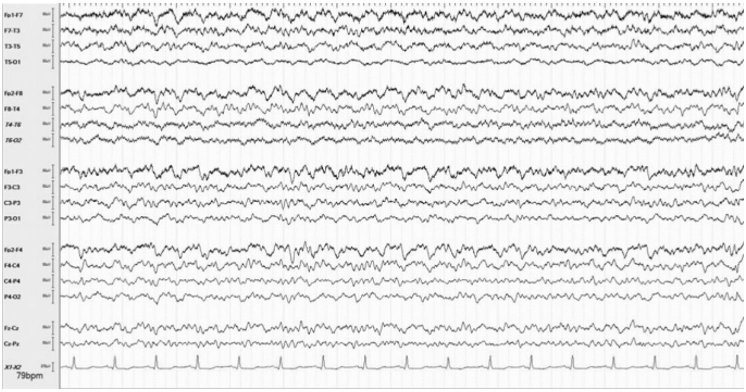

Figure 25.5 Fifty-seven-year-old man with cardiac arrest, generalized rhythmic delta with low amplitude sharps (GRDA plus S).

3. ***Benign EEG***: No malignant features. This may predict a favorable outcome [3].

Seizures/Status Epilepticus Post Cardiac Arrest

Seizures and status epilepticus occur in about a third of patients who are comatose after cardiac arrest. They are typically nonconvulsive requiring

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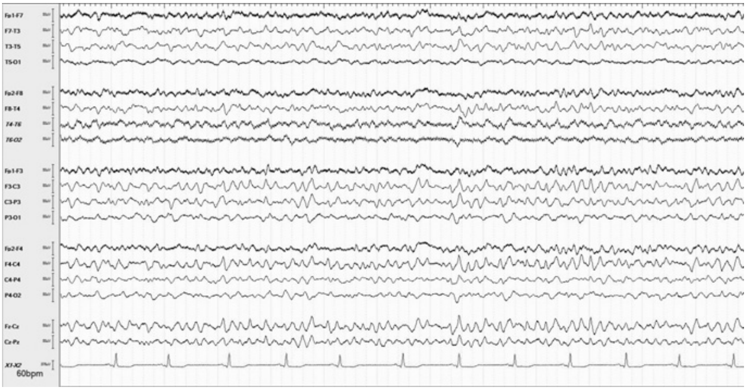

Figure 25.6 Fifty-four-year-old man with cardiac arrest, alpha-theta pattern with reversal of the anterior–posterior gradient.

continuous EEG monitoring for diagnosis. Seizures may be focal, generalized or multifocal in onset and tend to occur early during hospitalization or during rewarming [5]. Figure 25.7 shows NCSE post cardiac arrest. Status epilepticus post cardiac arrest is independently associated with increased mortality and continuous ictal activity may contribute to poor outcome [6].

Myoclonic SE is characterized by prolonged repetitive generalized or multifocal myoclonic jerking (often 30 min or more) that is time locked with bursts of burst-suppression (Figure 25.3) or associated with generalized periodic discharges (Figure 25.4). Early myoclonic SE (within 48 hours) from return of spontaneous circulation is highly predictive of poor outcome, though rare neurological recovery has been reported. The EEG is useful to identify reactivity and other signs of recovery in those with postarrest myoclonic SE after tapering off sedation [7].

Chapter Summary

1. EEG is useful to make prognostic estimations and evaluate for nonconvulsive seizures post cardiac arrest.
2. Unfavorable EEG patterns portend poor outcomes, whereas early improvements including restoration of physiological rhythms is associated with better outcomes.
3. The EEG has limitations; it should not be used in isolation to make prognostic estimations.

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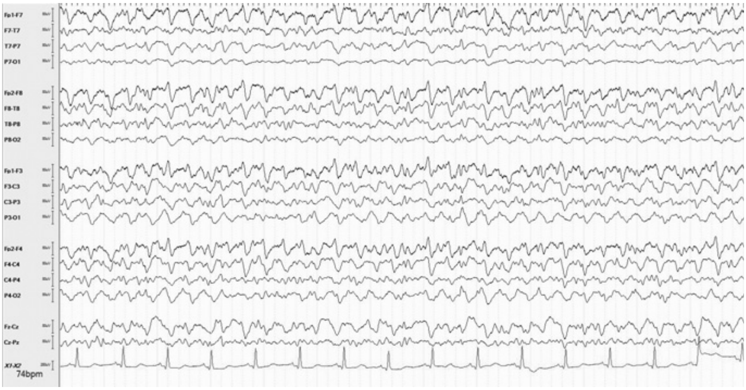

Figure 25.7 Same patient as in Figure 25.5, evolving generalized rhythmic sharp discharges consistent with NCSE.

4. Highly malignant EEG patterns are specifically associated with poor outcomes.
5. Individual malignant patterns have low sensitivity and specificity for poor outcomes.
6. Benign patterns (absence of malignant features) suggest favorable outcomes.
7. Seizures and status epilepticus are common post cardiac arrest and may contribute to poor outcomes.
8. Myoclonic status epilepticus is characterized by prolonged cortical myoclonus. Early myoclonic status epilepticus post cardiac arrest is typically associated with poor prognosis.

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